Quantisation, Serving and Harness Repair

What It Takes to Run a Local Coder Beside Real Work

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Abstract

The second night left a local worker that could finish small fixes once the serving stack was repaired and the prompt named the file; it left open what else separates a local model from a cloud worker. This third night narrows the question with eight candidates that ran fresh arms (two more were rejected before grading), paired precise and plan prompts on the same replay tasks, and one self-directed queue that ran to the morning. Four findings stand out. First, quantisation decides, and the lift replicates across model families: Ornith-35B-A3B at 8 bit instead of 4 bit turned per-task instability into two concentrated losses (5 of 8 graded arms against 7 of 17, Wilson 31–86 %) and produced the first local double pass on box-press; on a second family the same lift went from 4 of 6 to 5 of 6 on the identical three-task protocol, at half the resident size (17–18 GB against 35.5). Second, the remaining failure classes are model-side, and one is harness-repairable: dotted tool names (`Glob.glob`, `Read.read`, `Bash.Bash`) caused all 27 of the night's dotted-name errors, every one in arms spawned before the repair hook was registered at 19:39 — a 38-minute registration gap that the failing arms straddled; invented absolute paths caused 130 path-guard events (97 repaired, 33 denied; 17 distinct invented roots). Third, speed without instruction-following is worthless: a dense 30B model with the fastest streams of the night (25–29 tokens/s single stream) passed 0 of 5 graded arms. Fourth, a 125B MoE paged from SSD is usable (3 of 6, swap flat) but every decode token pays random expert-row page-ins — 3.0 tokens/s against 9.8 resident — and an 80B 8-bit model on the same host never went resident (10.7 GB rss) and never hit the prefix cache (0 hits in 102 fetches). The save-model-ref class remains uncrossed (0 of 14 graded local tries), and its replication attempt wedged the serving process twice (Metal 46.2 GB active against a 34.6 GB limit, resident collapsed to 0.1 GB). A morning head-to-head then settled the disputed rank-1 seat on the log-size tiebreak: the 30B-A3B 8-bit passes 4 of 4 gradeable log-size arms against the 35B's 3 of 4, at 3.8–4.4× the wall speed and roughly 31 GB resident, and takes rank 1 — a diff-verified rank 1, not a build-verified one: the 35B keeps the night's only build-verified passes, and a daylight build pass may still flip the seat. The recommendation: the 30B-A3B 8-bit as resident coder, the 35B 8-bit as build-verified fallback, multi-file saves on the cloud coder.

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1 Introduction

Night 1 asked whether open-weight models on one workstation can act as agentic coding workers and answered no for real fixes (Anthropic, 2026a). Night 2 asked the same question with paired arms and found that the answer was being suppressed twice: the serving stack never hit its prefix cache, and the worker mistyped paths until a repair hook was added; with both fixed, a local model finished one-function fixes within about twice the time of the cloud reference. The question of this third night is narrower still: *with serving and path repair solved, what else separates a local coder from a cloud coder?* Four candidate answers were put to the test, one per finding:

- ▷ **Quantisation.** The MoE candidate of night 2 ran at 4 bit and was unstable per task. The same model at 8 bit (Ornith-1.5-35B-A3B, "ornith8") is tonight's first hypothesis: the bit width, not the model, decides whether the per-task results hold together. When the queue extended in the morning, the same question was asked a second time on a different family (Qwen3-Coder-30B-A3B, 4 bit on night 1 against 8 bit tonight).
- ▷ **Model-side failure classes.** The failures that remain after the serving and path fixes — dotted tool names, invented absolute paths, loops — should now be attributable to the models, and the dotted-name class should be repairable in the harness, measured live.
- ▷ **Instruction-following, not speed.** A new dense 30B candidate (Muse-Glimmer, 4 bit) with the fastest streams of the night tests whether speed alone moves any task.
- ▷ **The uncrossed class.** The multi-file, reference-preserving save-model-ref task should remain beyond every local candidate, as it was on night 2, and stay with the cloud coder.

After the first two candidates and the serving fault of Section 5.8, the night turned into a self-directed queue that ran to the 08:15 close and served four more candidates, two of them experiments in where a model is allowed to live: a 125B MoE deliberately larger than free memory, paged from SSD, and an 80B 8-bit model that the operating system keeps out of residence entirely — and closed with a log-size head-to-head that settled the rank-1 seat (Section 5.2). Section 2 describes what changed since night 2, Section 3 the setting, Section 4 the method, Section 5 the results, Section 6 the harness findings, and Section 7 the relation to published work.

2 What Changed Since Night 2

The 8-bit arm. The quantisation A/B of Section 5.3: the same model family and launcher as night 2, at 8 bit instead of 4 bit. The 4-bit model failed its smoke test on night 2 (an assertion error on the one-line bug); the 8-bit model passed it.

Tool-name repair, measured live. Local models hallucinate namespaced tool names — `Glob.glob`, `Read.read`, `Bash.Bash` — and every such call dies with "no such tool available" as a tool-result error that no hook saw, so the run thrashes on the same mistake (the same unavailable name sixteen times in one arm tonight). A worker-only PreToolUse hook now rewrites a dotted name whose last segment names a known tool, case-insensitively, to the bare name and keeps the input; each repair is logged. The rule and the hook were committed at 19:01 and the hook was registered live at 19:39; the gap and its lesson are Section 6.2.

Finisher discipline. Night 3 lost arms to phase budgets, not to models: the first ornith8 phase was cut by the phase worker's deadline at 18:29 (four arms void, marked not-started), and a second worker deadline at 19:13 orphaned two glimmer arms mid-run. The rule adopted afterwards: when a phase is cut, the missing task–prompt cells are re-run as one completion phase per candidate, paired arms started together, one server loaded at a time (glimmer's server stopped 20:02, ornith8 loaded 20:03, stopped 21:33), and the void rows stay in the table as history instead of being quietly dropped.

An ATEM tool parser for Glimmer. The Muse-Glimmer template carries its own tool-call template (`atem:function_calls`); no shipped parser of the serving stack, including `auto`, reads it. A local parser was written into the patched launcher and probe-verified against all 38 choices on a sample before the server was started. The template raises on string tool arguments, but the serving layer's template-safety normalisation already converts them, and no HTTP 500 occurred all night.

The queue. After the wedge, the operator extended the night: a worker ran the candidate queue self-directed until morning — the box-press debt run owed by qwen38, gpt-oss-20b, the SSD-paged 125B (Section 5.6), a 6-bit Ornith completion, the two 8-bit builds of Section 5.1, and — at the 08:15 close — the log-size head-to-head that settled rank 1 (Section 5.2). Every candidate was smoke-tested (one-line bug through the full tool loop) before any task arm; four of the six new servers passed the smoke in 19–68 s.

3 Setting

All measurements ran on the same Apple MacBook Pro with an M5 Max and 128 GB of unified memory as nights 1 and 2, set to maximum performance, with the load average logged per arm. Local models were served with vllm-mlx (vllm-mlx contributors, 2026) through patched launchers (thinking off where the template supports it, output capped at 8192), one model server at a time. Workers were started through PAI (Nott, 2026). The workloads are anonymised as in the earlier reports: Project A is the native PDF application that supplies the coding tasks.

One constraint is carried over from the reboot of night 2 and tightened. A *36 GB watch line* bounds the resident size of the one allowed server, measured as resident memory of the serving

Table 1: *Models of the third night (night-2 candidates carry their rows into Table 2; the cloud yardsticks of night 2 are cited there, not re-run).*

Model	Placement	Quant	Role
Ornith-1.5-35B-A3B	local, MoE (3B active)	8-bit	Quant A/B arm, resident pick (“ornith8”; 35 G on disk)
Muse-Glimmer-30B	local, dense	4-bit	Candidate, rejected (“glimmer”; vision section present, text path forced)
Qwen3-Coder-30B-A3B (Qwen Team, 2025)	local, MoE (8 active)	8-bit	Queue candidate (“local30x8”; 32 G on disk)
Qwen3.8-Flash-Next	local, MoE (125B trunk), SSD-paged	3.3 bpw mixed	SSD-overrun experiment (“ddalcu”; 80 G on disk)
Qwen3-Coder-Next 80B	local, MoE (512E/10 active), mmap	8-bit	Queue candidate (“codernext”; 80 G, never resident)
gpt-oss-20b	local, MoE (36 experts)	MXFP4-Q8	Rejected at smoke (“gpt20b”; 11 G on disk)
Ornith-1.5-35B-A3B	local, MoE	6-bit	Queue completion (“ornith6”)
Qwen3.8-27B	local, dense	6-bit	Night-2 candidate; box-press debt run
GLM-4.7-Flash (Z.ai, 2026)	local	6-bit	Night-2 candidate, carries its ranking row
GLM-5.2-colibri E8-IQ3	local (never served)	E8-IQ3 3.06 bpw	Round-7 reject before serve: proprietary E8-lattice container format no local runtime dequantises; 289 G on disk
Qwen3-Coder-480B-A35B, small MLX quants	local (never served)	4-bit	Round-7 skip: no published quant fits free RAM (∼250–270 G on disk against ∼90 G free)
Claude Sonnet 5.5 (Anthropic, 2026b)	cloud	–	Yardstick (night-2 numbers, cited)
GLM-5.3-flash	cloud	–	Yardstick (night-2 numbers, cited)

process, not requested: ornith8 was measured at 35.0 GB at smoke start and 35.29 GB at the cut-off of its first phase, and peaked at 35.5 GB during the completion phase — half a gigabyte under the line, with the Metal allocation limit set to 34.6 GB (30 % of 115.4 GB). Glimmer, dense at 4 bit, sat far below it (15.5–16.0 GB resident, Metal peak 17.3 GB), and local30x8 lower still (17.2–18.1 GB), which is itself data for the resident-coder question: the 30B class is the only one that leaves the machine headroom. What the watch line does not prevent is Section 5.8: Metal activity can exceed the allocation limit from outside the cache, and that is where the night's serving fault lives.

4 Method

4.1 Tasks, prompts, grading

The four replay tasks of nights 1 and 2 are reused unchanged: fixes that already landed in Project A, started from the parent commit in an isolated clone, graded by the hidden tests that came with the fix. Deadlines are 20 minutes for view-decimals, 15 for log-size, 25 for box-press and 40 for save-model-ref (30 minutes on the SSD-paged arms, where every arm was expected to run to its cap); wall time is censored at the deadline and the build and test exit codes are recorded. Each task runs as a pair of arms started together, the *precise* prompt (file and function named) against the *plan* prompt (a cloud-written edit plan, "cloud head, local hands"). A patch passes when the result builds and the hidden tests pass; no model judges another. Wilson intervals (Wilson, 1927) are given for pass rates.

4.2 Two grading modes, and when each counts

The night used both graders, and the ranking states which row rests on which. Through the ornith6 completion (23:37) every verdict is a *build verdict*: the hidden tests ran, and pass means the tests went green — these are the same verdicts as nights 1 and 2. In the later queue rounds the builder was dropped (the operator banned the heavy build as a watch-line risk), and arms were graded by diff comparison against the landed fix: *check* when the arm touched the fix's own file and function, *likely-fail* otherwise. Checks were then inspected by hand; every pass quoted for local30x8, ddalcu and codernext is an inspected diff that contains the true fix (for log-size, the exact one-line root cause), and Section 10 carries the cost of this substitution. Build verdicts and inspected checks are never mixed inside one fraction without a note.

4.3 Replication rule

Night 2's rule "no single-run verdicts" is enforced as a replication pass: every cell of the shortlisted candidate is run a second time (rep 2) before the ranking is called final. Tonight the replication pass is incomplete by a serving fault, not by choice: view-decimals replicated (the split repeated exactly), the save-model-ref pair wedged the server twice (Section 5.8) and was abandoned, and log-size and box-press were not re-run — both had passed 2 of 2 in rep 1. The ranking therefore stands on rep-1 evidence with one replicated task, and the sample sizes are stated wherever a fraction is quoted.

4.4 Voids

An arm that never reached the model or was killed by the harness before grading is *void*, never a fail. Tonight's voids have three causes: phase-budget kills (six arms), a loop-guard kill on a degenerate arm (Section 5.7), and the wedged server (two arms).

5 Results

5.1 Ranking

Table 2: *Final night-3 ranking (post head-to-head, the 08:15 verified close; rep-1 evidence plus the log-size rep-2 arms of Section 5.2). "Result" is graded arms on the three-task set view-decimals/log-size/box-press; ornith8 additionally holds save-model-ref 0 of 2. Rows marked* build *are build-verified verdicts (Section 4.2);* check *rows are inspected diff-level passes — rank 1's seat is diff-verified, not build-verified. Cloud yardsticks from night 2 for orientation: Sonnet 5.5 passed 4 of 4 (median 5.6 min), GLM-5.3 and GLM-5.3-flash 3 of 4 each, Haiku 4.5 2 of 4.*

Rank	Candidate	Result	Resident	Decode	Note
1	Qwen3-Coder-30B-A3B 8-bit (local30x8)	**5/6** (view 2/2, log 2/2, box 1/2); log 4/4 across both reps	17–18 GB round 6; 30.9–32.4 GB h2h window	8.4 bulk, 62 max single; 13.8 avg h2h	*check*; inspected; takes rank 1 on the h2h (Section 5.2)
2	Ornith-1.5-35B-A3B 8-bit (ornith8)	**5/6** (box 2/2, log 2/2, view 1/2) + save-model-ref 0/2; log 3/4 with rep 2	35.5/36 GB	9.8–10.3 bulk, 22.8 s/turn; 8.8 avg h2h	*build*; rep-1 passes build-verified; first log-size precise FAIL in rep 2
3	Qwen3.8-Flash-Next 3.3 bpw (ddalcu)	3/6 (log 2/2, box 1/2 precise, view 0/2)	∼52 G of 128 G wired	3.0 bulk	*check*; SSD-paged, Section 5.6
4	Qwen3-Coder-Next 80B 8-bit (codernext)	2/6 (log 2/2; view, box read-only)	10.7 GB rss (80 G never resident)	2.5 bulk, 8.0 max	*check*; prefix cache never hits
5	GLM-4.7-Flash 6-bit (night 2)	5/7 (log 3/3, box 1/2, view 1/2); save-model-ref unfinished	∼24 GB	–	night-2 rows carried
6	Ornith-1.5-35B-A3B 6-bit (ornith6)	3/6 (box 2/2, log 1/2, view 0/2)	–	–	*build*; ranks between ornith8 and qwen38
7	Qwen3.8-27B 6-bit (qwen38)	3/8 (log 2/2, view 1/2, save-model-ref 0/2, box 0/2)	∼20 GB	–	box-press debt run tonight
8	Ornith-1.5-35B-A3B 4-bit (night 2)	7/17 (log 4/4, view 2/5, box 1/4, save-model-ref 0/4)	26 G disk	–	superseded by its 8-bit
–	Muse-Glimmer-30B 4-bit (glimmer)	0/5 graded (+3 void)	17.3 GB	14.9 (25.2 max)	rejected: instruction-following
–	gpt-oss-20b MXFP4-Q8 (gpt20b)	smoke fail ×3	–	–	rejected: no tool loop (Section 5.9)
–	Muse-Spark GGUF / qwen4_exp via llama.cpp	never served	–	–	llama-server 0.7.0 supports neither architecture

The head-to-head of Section 5.2 moves rank 1 to local30x8. Until it ran, ornith8 held the seat on evidence strength — the only candidate whose passes are build-verified, the only one that passes real tasks on both prompt styles, and the only one that passed box-press at all until local30x8's inspected check (night 2's local models passed it three times in total, each on one arm of a pair); its Wilson interval on the night's graded arms, 5 of 8, is 31–86 %. On footprint and arm speed the challenger dominates it: local30x8 passes the same five of six cells at 17–18 GB (half the resident size) with arms of 5–6.5 minutes against ornith8's 12–25, and its one round-6 failure (box-press plan) is the same cell its own night-1 4-bit build failed. The tiebreak broke the tie on both axes — 4 of 4 gradeable log-size arms against 3 of 4, at 3.8–4.4× the wall speed — with the tightest diff of the night's log-size arms (sim 0.80, ratio 0.67, smaller than the landed fix). What keeps the seat provisional is grading mode: rank 1 is diff-verified, not build-verified, and ornith8's rep-1 passes remain the night's only build-verified ones, so a daylight build pass may still flip the seat back. Ornith8's rep 2 also added its first log-size precise FAIL, on a task it had passed twice in rep 1 — variance at 8.8 tok/s is real. Ranks 3 and 4 are the memory experiments of Section 5.6. Qwen3.8 stays the light fallback at roughly 20 GB; its owed box-press run happened tonight and failed both arms with real effort (23 and 34 turns, no files touched), so its record is now complete at 3 of 8. Glimmer is rejected (Section 5.5), and the 4-bit ornith arm is superseded by its 8-bit self.

5.2 The rank-1 head-to-head

Round 7 resolved the queue's two remaining candidates before either served — a GLM-5.2-colibri pack rejected on model format (a proprietary E8-lattice container no local runtime dequantises, 289 G on disk against ∼90 G free, and its own README's reference host decodes at 1.57 tok/s) and Qwen3-Coder-480B small MLX quants skipped on the free-RAM rule (no published quant under ∼250 G on disk; the SSD-paging tax was already falsified twice at 80 G) — and spent the window on the rank-1 question instead. Both front-runners held 5 of 6 with ornith8 ahead on grading strength alone, so the tiebreak re-ran the one task both had double-passed, log-size, as a rep-2 pair per candidate: one server at a time (ornith8 on :8015, then local30x8 on :8034), the verify-up gate before each server, a memory watchdog live throughout (zero kills, swap 18.9 → 17.8 GB), 15-minute caps, buildless diff grading with every diff inspected.

Table 3: *The rank-1 head-to-head: log-size rep-2 arms, one server up at a time, 15-minute caps, buildless grading with diffs inspected. "Walls" are the arms' wall times.*

Arm (log-size, rep 2)	Verdict	Wall
ornith8 precise	likely-fail — cap, 0 files, no commit; its first log-size precise fail, on a task it passed twice in rep 1	15m01s
ornith8 plan	**pass** (check) — the landed fix's exact root cause (cached size key → `attributesOfItem`, never cached) plus a one-URL rotation test; committed, inspected	10m57s
local30x8 precise	**likely-pass** — committed `bd3ee3c0`, 2-line fix in `DebugLog.swift` + 45-line rotation test; files/functions hit 1.0, sim 0.80, ratio 0.67	**2m55s**
local30x8 plan	**pass** (check) — `attributesOfItem` at the fix's exact line plus an extended test, uncommitted	**2m30s**

Local30x8 wins on both axes: pass count, 2 of 2 against 1 of 2 in rep 2 (4 of 4 gradeable log-size arms across both reps against ornith8's 3 of 4), and walls, 3.8–4.4× faster on the same arms — both its arms self-ended at 2.5–3 minutes against a cap-out and an 11-minute finish. The server windows agree with the verdict: streams averaged 8.8 tok/s on ornith8 (n=22) against 13.8 on local30x8 (n=24, max 53.8), and both processes sat at 30.9–32.4 GB resident — the 30B's residency is memory-pressure-dependent (17–18 GB in round 6), so the ∼31 GB of the tiebreak window, not the round-6 figure, is the planning number. The verdict, verified against the grade rows at 08:15: rank 1 moves to Qwen3-Coder-30B-A3B 8-bit; ornith8 drops to rank 2, kept close by its build-verified rep-1 passes, and takes the seat back if the daylight build fails local30x8's committed diff. The tiebreak cost two skips on the record: colibri and the 480B were never measured, only rejected with reasons. Servers were stopped and ports verified clear at close; the fitness run is closed.

5.3 The quantisation A/B

Table 4: *The same MoE model at 4 bit (night 2) and 8 bit (night 3). Pass fractions are graded arms per task; n is 4–17 for the 4-bit rows and 2 per cell for the 8-bit rows.*

	Ornith 4-bit (night 2)	Ornith 8-bit (night 3)
Smoke (one-line bug, driver-verified test)	fail — assertion error	pass — bug fixed, test exit 0, 36 s wall
log-size	4/4	2/2
view-decimals	2/5	1/2 (precise timed out; plan passed)
box-press	1/4	**2/2** — first local double pass
save-model-ref	0/4	0/2
Graded total (Wilson 95 %)	7/17 (22–64 %)	5/8 (31–86 %)
Failure shape	per-task instability: every task fails in at least one arm	two concentrated losses: the view-decimals precise timeout and save-model-ref
Weights on disk / resident	26 G / (night-2 serving)	35 G / 35.5 GB peak
Bulk decode	52.8 tok/s single request, 0.8–1.5 s/turn after the cache fix	10.3 tok/s bulk (4229 tokens in 411.6 s); 22.8 s/turn on the passed plan arm
Prefix cache	fixed launcher, clean	clean: 158 hit / 7 miss (phase), 312 hit / 6 miss (completion)

The 8-bit arm answers the hypothesis in its favour with one caveat. What changed is not the average pass rate — 7 of 17 and 5 of 8 are the same result at different sample sizes — but the shape: the 4-bit arm failed every task in at least one arm, while the 8-bit arm fails only in two places, one of which (view-decimals precise) is a timeout at 22.8 s per turn rather than a wrong change. The caveat is speed: 8 bit halves nothing about the MoE's active parameters, but the measured bulk decode fell to 10.3 tok/s and the passed plan arm of view-decimals ran at 13.7 minutes and 36 turns (22.8 s per turn), so the resident model spends wall time the operator feels. Night 2's reading that "collapse below 30B-A3B" is about speed times quality, not size alone, is unchanged.

The lift replicates on a second family. The queue's last candidate is the same experiment on Qwen3-Coder-30B-A3B, which ran night 1 as a 4-bit LM Studio build ("local30", 4 of 6 on the three paired tasks). At mlx-community 8 bit, same protocol, the same three-task set goes to 5 of 6 (view-decimals 2/2, log-size 2/2, box-press 1/2): the same 30B-A3B MoE, 128 experts, uniform 8-bit group-64, at 17.2–18.1 GB resident. Every pass is an inspected diff-level check, and each contains the true fix: view-decimals precise committed a two-file fix in the fix's

own file and function (plain-decimal camera formatting) plus a test after about five minutes, early-stopped; both log-size arms found and committed the exact root cause (the cached file-size key read instead of the real bytes of the file); box-press precise is the fix's one-liner (the pick test that must exclude 3D annotations) plus a forty-line test, uncommitted at the cap. The two weak cells are honest: view-decimals plan sprawled over a 22-file uncommitted diff at the cap, and box-press plan touched nothing. One lift is a finding about one model; two lifts on two families, on the same paired protocol, is a finding about quantisation — and the second family delivers the passes at half the memory and two to four times the arm speed, which the morning head-to-head then confirmed on the rank-1 seat (Section 5.2).

5.4 Per-task splits

Table 5: *Ornith8 by task, prompt and repetition (minutes, turns where recorded; 20/15/25/40 min deadlines; lines are changed lines in the fix files against the landed fix).*

Task	Rep	Precise	Plan	Lines (arm/fix)
view-decimals	1	fail 20.0 (cap, build failed)	**pass** 13.7 / 36 turns	24 and 12 vs 9
view-decimals	2	fail 17.8 (built, test failed)	**pass** 20.0	11 and 12 vs 9
save-model-ref	1	fail 40.0 (cap, built, test failed)	fail 40.0 (cap, built, test failed)	0 of 3 files, 0/65
save-model-ref	2	void — server wedge	void — server wedge	–
log-size	1	**pass** 12.0 / 26 turns (7042 tok)	**pass** 15.0	2/3 and 8/3
box-press	1	**pass** 25.0 (cap)	**pass** 25.0 (cap)	6/5 and 40/5 (3 extra files)

The box-press plan arm made the change three times too broad (40 lines against 5, three extra files) but the viewer it was to open works; the hidden tests pass. The phase rows of 17:49–18:29 are the first pass; the completion phase (20:03–21:33) re-ran the cells the deadline kill had orphaned; rep 2 started 21:39.

Two regularities carry over from night 2. The plan prompt again wins view-decimals for ornith8 (2 of 2 graded arms, against 0 of 2 precise, across two repetitions — the night-2 pattern of five local plan passes on that task with no precise pass now has six data points), and save-model-ref again admits no local pass: built, tested, wrong behaviour in both arms, 102 and 144 turns respectively, no diff in the three fix files. What is new is box-press: both arms pass, one of them (precise, 6 lines) tightly. The pattern has one exception at 6 bit: ornith6 failed view-decimals on *both* arms — the first local plan-prompt failure on that task — while passing box-press twice, so the plan-prompt regularity is not quant-independent.

Table 6: *The queue candidates by task (precise / plan; deadlines 12–30 min as stated in Section 4.2; "check" = inspected diff-level pass, "l-fail" = likely-fail).*

Candidate (cap)	view-decimals	log-size	box-press	Night total
local30x8 (12/15/15)	**pass** (check): committed 2-file fix, ~5 min early-stop / sprawl: 22-file uncommitted diff at cap	**pass** (check) ×2: exact root cause, both committed (plan arm also committed scratch files)	**pass** (check): the 1-line pick fix + 40-line test, uncommitted at cap / l-fail, 0 files	**5/6**
ddalcu, SSD-paged (30)	fail: 45 turns, 0 files / fail: 36 turns, test file only	**pass** (check) ×2: the rotating-log fix, committed / from the tree	**pass** (check): the pick fix's core, 4 lines vs 5 / fail: 36 turns, 0 files	**3/6**
codernext (12/25/25)	fail: 34 turns, 0 files / fail: 31 turns, 0 files	**pass** (check) ×2: the exact 1-line fix, uncommitted, self-reported fail (no build in cap)	fail: 0 files / fail: wrong file only	**2/6**
qwen38, box-press debt (25)	–		l-fail: 23 turns, 0 files / l-fail: 34 turns, 0 files	0/2
ornith6 (20/15/25, *build*)	fail 0.2 / fail 13.4, 79 turns	fail 0.4, 1 turn / **pass** 15.0	**pass** 25.0, 8/5 / **pass** 25.0, 66/5 (3 extra files)	**3/6**

Ornith6's view-decimals and log-size precise arms are degenerate one-turn runs (0.2 and 0.4 minutes; the log-size arm emitted 189 tokens and stopped) — the night-2 "empty first turn" class resurfacing at 6 bit. The codernext log-size arms produced the true fix but the worker judged itself failed because it could not run the build; the diff was graded instead.

5.5 Glimmer: fast, useless

Table 7: *Muse-Glimmer-30B 4-bit by task (dense 30B; all arms on its own server, :8016).*

Task	Precise	Plan
view-decimals	fail 20.1 (cap, 0 files changed)	fail 20.0 (cap, 0 files changed)
save-model-ref	void (deadline kill, 62 turns deep)	void (deadline kill)
log-size	fail 15.0 (29 tool calls, all read-only)	fail 2.4 / 1 turn (answered the protocol, not the task)
box-press	fail 25.0 (cap, ∼120 turns, 0 edits)	void (loop guard, 2.3 min)

Glimmer is the night's control on the speed hypothesis, and it fails cleanly. The serving stack was not the problem: first token in 0.7–1.2 s, 25–29 tok/s single stream (83–141 tokens per turn in 2.9–4.9 s), prefix cache 194 hit / 7 miss in the phase and 124 hit / 4 miss in the completion run. The worker explored correctly and never landed: roughly 69 turns per arm on view-decimals with no edit, 29 read-only tool calls on log-size, about 120 turns on box-press with zero edits, against ornith8's 36-turn pass on the same task class. Three distinct model-level failure modes appear in its five graded fails: *read-only paralysis* (explores to the cap and never edits), *protocol confusion* (a one-turn run that answered the worker protocol's handoff message instead of the task, 1492 tokens, no tool call), and the *tool-name loop* of Section 5.7. The verdict of the ranking — a fast talker and a slow coder — doubles as a positive result: a dense 30B at 4 bit is fast enough on this machine (2.5–3× ornith8's single-stream decode), so density and quantisation are no longer reasons to reject a candidate; instruction-following in the agent loop is the gate.

5.6 The SSD-overrun experiment

The queue's most interesting question was not "which model" but "where may a model live". Two candidates deliberately violate the resident-size assumption. The first, "ddalcu", is a 125B qwen4_exp MoE (125B trunk, 51B n-gram table, 4B MTP head) packed at 3.3 effective bits-per-weight — below the ∼3.5-bit threshold at which on-device evaluations start to bend (*A Systematic Evaluation of On-Device LLMs: Quantization, Performance, and Resources* 2025) — 80 G on disk against free RAM; its n-gram table is mmapped from an external store and never resident, so the server *must* page weights from SSD during decode. Three load blockers were found and fixed in sequence before it served at all: an MTP head stored under an unexpected key (stripped; served without MTP speculation); a mixed 2/3/4/8-bit pack whose config declares only a global 4 bit (a generated quantisation map of 774 per-tensor entries, proven by a strict lazy load of all 3167 tensors); and folded RMSNorm weights that turn $(1+w)\cdot x$ into $(2+w)\cdot x$ under the serving stack's formula (monkeypatched to the plain form; the arithmetic sanity check "17 × 23" then answered "391"). After that: smoke pass through the full tool loop in 68 s, zero template errors, 3 of 6 graded arms with log-size on both prompts and the box-press precise fix's core — the second-best quality-per-task-attempted of the night.

The cost is the finding. Over 202 streamed turns, decode averaged 3.0 tok/s (median 3.3, min 0.7, max 14.2); the machine survived with swap growing 25 MB in total against a kill line of 500 MB per five minutes, and the server peaked at 32.7 GB rss (∼51–57 G of 128 G including Metal). But the paging tax is paid on *every* token, not only cold turns: the first big prefill ran as interleaved SSD reads (1,307 slow-step warnings), and warm turns with a fully cached prefix still decoded at 3–4 tok/s because MoE decode pages random expert rows. Every arm ran its full 30-minute cap; at 3 tok/s that is about 60 turns — enough for log-size and box-press, never enough for view-decimals to converge. Speed, not capability, is the SSD-overrun bottleneck: the same arms that ornith8 finishes at 9.8 tok/s take a 3× wall-clock tax here.

The second experiment, "codernext", is the 80B hybrid-attention MoE at 8 bit (512 experts, 10 active) on the same host — and the operating system solved residency its own way: the server process sat at 10.0–10.8 GB rss the whole run, the 80 G of weights were never resident, and macOS paged expert rows from the file cache on demand. The load was harmless (26–28 % free, swap fell 20.0 to 19.1 G, the guard never fired; loading displaced ∼19 G of other sessions' memory into swap first), the smoke passed in 59 s — and the model managed only 2 of 6, all four view-decimals and box-press arms read-only to the cap at 2.5 tok/s (about 2 turns per minute), while its two log-size arms again contain the exact one-line fix. It also produced the night's cleanest cache result: **zero prefix-cache hits in 102 fetches** over 205 requests (against 166 hits in 2 misses for the qwen3_moe family the same night on the same launcher), so every turn re-prefills the whole prompt. Hybrid linear-attention architectures appear to have no working prefix-cache path in this serving stack; until they have, an 80G resident-free coder pays full prefill on every turn, and the "pages little at mixed quantisation" hope is falsified at pure 8 bit.

5.7 Failure taxonomy

The taxonomy is deliberately split by owner. The dotted-name class is mechanical and was repaired mid-night (Section 6); the invented-root class is model-side but already contained by night 2's path hook, which repaired 97 and denied 33 invented paths and kept every arm inside its clone — the repairs are themselves evidence of what the models believe a path looks like (a home directory named after someone else, a `/workspace` root, a bare `/Sources` where the clone's tree should begin); the wedge is serving-side and new. Glimmer's behavioural classes have no lever yet, and the class generalises: read-only paralysis recurs in every candidate below about 3 tok/s, where the deadline fills with reading before the edit can come — at SSD-paged speeds the failure mode is a property of the wall clock as much as of the model.

5.8 The serving wedge

At about 22:11, one third into the save-model-ref rep2 pair, the ornith8 server stopped inferring: Metal reported 46.2 GB active with 50.8 GB peak against the 34.6 GB allocation limit and an empty (0.0 GB) cache, the serving process's resident memory collapsed to 0.1 GB, and the disconnect guard spun — one client connection was held open past 700 s, a second

Table 8: *Failure classes of the night with counts from the tool-error and path-repair logs (experiment arms only; 136 error rows in the night's window, a further 175 coming from an out-of-round infrastructure session on the same machine).*

Class	Count	Where and what
Dotted tool names	27 rows, 2 arms	All glimmer, all in arms spawned before 19:39: view-decimals pair 10 (`Glob.glob` 3, `Read.read` 4, `Bash.Bash` 3); box-press plan 17 (`Glob.glob` 16, `Bash.Bash` 1), killed by the loop guard after 2.3 min. The ten arms spawned after registration produced none.
Invented absolute paths	130 guard events	97 repaired (the suffix of the invented path existed under the clone), 33 denied. 17 distinct invented roots; most common `/Users/vv` (23), `/Users/cheric` (13), `/Users/ab` and `/home/ubuntu` (8 each), the `/Users/ele*` family (16), `/workspace/src` (6), `/Users/slyvar` (6). By task: save-model-ref 26, box-press 40, view-decimals 23, log-size 8. By model: ornith6 36, ornith8 34, glimmer 26.
Server wedge	2 events	Both on the ornith8 rep2 server during save-model-ref (Section 5.8).
Read-only paralysis	8 arms	Glimmer: view-decimals ×2, log-size precise, box-press precise. codernext: view-decimals ×2, box-press precise (the whole cap at ∼2 turns/min). ddalcu: view-decimals precise. Right files, no edit landed — the cap, not the search, is the limit at 2.5–15 tok/s.
Protocol confusion	1 arm	Glimmer log-size plan: 1 turn, answered the worker protocol's handoff message, 1492 tokens. (Two further one-turn ornith6 arms, Table 6, look like the same class but were not inspected.)

past 115 s — with two requests running and none completing. Both arms were killed by the arm guard and marked void. The wedge had occurred once before on the same server, identically, which makes it systematic in the night's sense: long-context arms (save-model-ref has the largest prompt of the four tasks) trip a vllm-mlx memory path that deadlocks after roughly 35–45 minutes under continuous batching. It is the third consecutive night whose most expensive finding is a serving fault invisible in any pass/fail table: night 1's serial prefill, night 2's silent cache defeats, tonight's deadlock. The scheduler's own Metal log line (active 46.2 GB) is the diagnostic to watch on future long-context phases; a per-phase Metal-peak watch line is the cheap lever, and the replication rule absorbed the damage — the ranking stands on rep 1, and the wedge is the finding. The queue adopted the consequence directly: no later candidate was run on save-model-ref (cloud class, wedge risk).

5.9 Serving findings

Table 9: *Server-side figures per phase (prefix-cache hit/miss from the server log; stream average and maximum decode tokens/s).*

Phase (server)	Requests	Cache hit/miss	tok/s avg (max)
ornith8, phase 1 (:8015)	–	158 / 7	10.3 bulk
ornith8, completion (:8015)	313	312 / 6	9.8 (15.8)
glimmer, phase 1 (:8016)	201 fetches	194 / 7	11.8 batched; 25–29 sin
glimmer, completion (:8016)	124	124 / 4	14.9 (25.2)
ornith6 (:8013)	–	–	–
local30x8 (:8034)	2 arms batched	166 / 2	8.4 bulk (62.0); 18.8 on the view-de
ddalcu (:8032)	202 streamed turns	warm prefills cached	3.0 (14.2), median 3.3
codernext (:8035)	205	**0 / 102 fetches**	2.5 final (8.0)

The night-2 cache fixes hold for every architecture they were built for: on ornith8, glimmer and local30x8 the only misses are the first fill of each arm, and the last ornith8 turn of the completion phase read 86,212 of its 86,641 prompt tokens from cache. No HTTP 500 occurred on any server; the 500-template failure of the previous serving attempt did not recur. Three further serving facts belong on the record:

A dead port. The gpt20b provider was registered on :8021, where a root-owned launchd socket accepts every connection and immediately resets it; the server could never bind (address-in-use twice). The provider was re-registered on :8031 and the port should be struck from the cockpit. A port that "accepts and resets" reads as a server that is up — the verify-up probe (models endpoint plus a one-token completion) is what caught it, and is now standard before any arm.

Mid-session system messages. The night-2 Qwen3.8 launcher disabled the serving stack's system-message merge because their templates reject a system message after the first. Claude Code appends system-role items mid-session, and with the merge disabled the

template raised "System message must be at the beginning." — 32 stream failures in the first 13 minutes of the qwen38 box-press phase. The new launcher keeps the merge (after which a probe with a mid-session system message returns a clean tool call); the OpenAI-protocol path has no merge and still rejects such histories, but the client never uses that path.

Harmony without the final channel. gpt-oss-20b answered every smoke with an empty message (*empty-response*, three times, about 12 s each). Without its harmony prompt rendering the model freehands tool JSON into plain text with no structured tool calls; *with* harmony installed and verified active, it emits only the analysis channel and stops (finish reason `stop` at 42 tokens; every delta is reasoning content, none is content or tool call). The MLX build never produces the final-channel choreography the client needs, so the model cannot drive the tool loop on this stack at all — rejected without a task arm. Raw non-tool prompts answer fine.

6 Harness Findings

6.1 Dotted tool names, repaired and measured

The class: a local model writes `Glob.glob` or `Read.read` — namespaced as if tools were modules — the client reports "no such tool available", and the model retries the same call, in the worst case sixteen times, until a loop guard kills the arm. The repair: a worker-only PreToolUse hook that, when the tool name contains a dot and its last segment names a known tool case-insensitively, rewrites the call to the bare name and keeps the input, logging each rewrite; anything else passes untouched. Measured live tonight: the repair log holds three rewrites (the registration probe `Glob.glob` → `Glob` at 19:38:54, and two rewrites in the very worker that shipped the hook), all 27 dotted-name error rows of the night sit in the two glimmer arms spawned before 19:39, and the ten arms spawned after it — including every ornith8 completion and rep2 arm — produced none. The class is closed at the cost of one hook and one log line per rewrite.

6.2 The settings-registration gap

The hook existed for 38 minutes before it fired: committed at 19:01, registered in the live settings at 19:39, and hooks are read when a worker spawns. The two box-press arms launched at 19:37 to close glimmer's grid therefore ran without it and hit the loop the hook exists for; both count as fails or voids in the tables above. The lesson generalises beyond this hook: *a harness fix is not deployed until the next spawn after its registration*, so mid-phase fixes must either wait for the phase boundary or be paired with a re-spawn, and any "the hook did not help" claim needs a timestamp check against the repair log before it is believed. The probe rewrites (`worker=probe` in the log) are the cheap proof of liveness; they should be a standard step of hook rollout, not an afterthought.

6.3 Finisher discipline

Seven of the night's voids are self-inflicted: six arms were orphaned by phase-worker deadline kills (18:28–18:29 on ornith8, 19:13 on glimmer) and one by the loop guard. None of the seven measures the model. The discipline adopted after the second kill — completion phases per candidate, paired arms started together, one server loaded at a time across the boundary (glimmer stopped 20:02, ornith8 loaded 20:03), voids kept visible in the table — recovered every orphaned cell within two hours, which is why the ranking covers all four tasks at all. The budget–arms interplay is the finding: a phase budget sized for the full grid will always spend its last minutes killing exactly the arms that a completion phase then has to redo. One interference of the opposite kind occurred in the morning rounds: both box-press arms of local30x8 were killed from outside the round 50 s in by another session's pane activity, with 13 minutes of cap left and no watchdog or deadline involvement; the pair was re-run clean and completed, the killed attempt's rows stay in the log as history, and the authoritative rows are the re-run.

7 Related Work

The series' tasks grade agents on real repository fixes with hidden tests, which follows the SWE-bench format (Jimenez et al., 2024); the harness levers measured tonight (tool-name repair, path denial, loop guard) are agent-computer-interface interventions in the sense of Yang et al. (2024), and the precise and plan prompts are external localisation, the constraint that Xia et al. (2024) identify as binding. The PAI worker layer is the open-harness counterpart of Wang et al. (2025). Everything serving-side rests on paged attention and prefix reuse (Kwon et al., 2023); the radix-tree reuse that SGLang formalises (Zheng et al., 2024) is exactly the mechanism that the night-2 cache defeats broke silently and that codernext shows missing *by architecture*: a hybrid linear-attention model on which the prefix cache never fires at all.

On quantisation, tonight's A/B is the agentic counterpart of the accuracy–performance trade-off studies: 8 bit behaves like the near-lossless regime of Kurtic et al. (2024) — failures concentrate instead of spreading — while the low-bit degradation literature (Li et al., 2025) locates failures early in reasoning chains, which matches the 4-bit arm's per-task instability; that quantised models are not uniformly weaker (Fang et al., 2025) is why the arms are paired, and the replication of the lift on a second family is the paired answer. On-device evaluations that co-report capability with system figures (*A Systematic Evaluation of On-Device LLMs: Quantization, Performance, and Resources* 2025) are the single-turn analogue of Section 5.6; the tool-calling angle separates this setting from function-calling benchmarks (J. Chen, Xiao, et al., 2024; Patil et al., 2024) — a dotted or namespaced tool name costs minutes of thrash inside a build-test-fix loop, not one point, and the trajectory-level failures (read-only paralysis, protocol confusion) are invisible to single-turn scoring. Reporting minutes and turns next to pass rates follows the cost-accounting discipline of Kapoor et al. (2024); on this night the cost axis is gigabytes and seconds, and it changes the ranking's conclusion (the cheaper candidate takes rank 1 on the head-to-head tiebreak).

The failure taxonomy maps onto the multi-agent failure analyses: MAST's verification and specification categories (Cemri et al., 2025) cover glimmer's read-only paralysis and protocol confusion at the single-worker level; the no-op-edit loops of night 2 and tonight's sixteen-fold retry are the "cognitive deadlocks" of Liu et al. (2025); infinite-loop analyses find such loops in the agent–harness feedback path rather than the model alone (Hou et al., 2026), which is where the loop guard and the repair hook operate; and the attribution dispute — users propose guardrails, developers blame the model — is exactly the split of Section 5.7 into harness-repairable and model-side classes (J. Chen, S. Chen, et al., 2026). What the published work on quantisation and tool use does not contain is the setting itself: quantised candidates graded by builds and hidden tests, paired arms on one GPU beside normal work, with the serving defects reported as first-class results.

8 Discussion

A resident coder exists now, with two candidates for the seat. Ornith-1.5-35B-A3B at 8 bit passes three of the four task classes on both prompt styles, replicates the view-decimals plan pass, and was the first local model through box-press on both arms; it holds the night's only build-verified passes. Its two caveats are exactly what the head-to-head weighed: 35.5 of 36 GB resident leaves no room for a second model server, ever, on this machine — the resident model is a commitment — and 22.8 s per turn is twice the operator's patience for a one-line fix, which the 12–15 min log-size passes make concrete.

Half the memory passes the same tasks — and now holds rank 1. The second family's quant lift is the practically important half of the quantisation finding: local30x8 matches ornith8's 5-of-6 cells at 17–18 GB resident and finishes its passes in 5–6.5 minutes against 12–25, and the log-size head-to-head then broke the tie on both axes (4 of 4 gradeable against 3 of 4, at 3.8–4.4× the walls). The seat stays provisional in one respect: rank 1 is diff-verified, not build-verified. If the daylight build passes, the resident-coder deal of the night is the smaller model, and the 35B arm's crown was an artefact of grading strength rather than of capability; if it fails, ornith8's build-verified record takes the seat back.

Memory beyond the machine is a wall-clock problem. The SSD experiments answer a question the resident debate usually blurs: a 125B model that does not fit can still be the second-best coder of the night on quality per task attempted — with three load fixes, 80 G of weights served productively from SSD, swap flat — but every decode token pays random expert-row page-ins, and at 3 tok/s every 30-minute cap fills with reading. The 80B variant shows the same tax with the weights never resident at all. The conclusion is not "bigger is possible" but "bigger needs RAM": the SSD-paged model is the likely new rank 1 on a host where it fits (128 G-class, or a smaller pack), and on this host it is a research instrument, not a worker.

The cloud keeps the multi-file class. save-model-ref (3 files, +64 lines) has now been attempted by every local candidate and passed by none: 0 of 12 graded local arms on night 2,

0 of 2 tonight, with the rep2 attempt voided by the wedge — 14 graded tries in total (Wilson 0–22 %). Sonnet 5.5 passes it in 7.0 minutes. The division of labour stands: local resident for one- and two-function fixes on both prompt styles, cloud for reference-preserving multi-file work.

Instruction-following is the gate, not the weights. Glimmer had the best streams of the night and the worst table; the dense-30B architecture is thereby acquitted on speed, and the next candidate queue should be ordered by agentic instruction-following evidence, not by tokens per second or density. The three glimmer failure modes are also a warning about evaluation: a single-turn codegen score would have ranked it first.

Replication rule earns its keep twice. The view-decimals split replicated exactly (precise fail, plan pass, both repetitions), which is what makes the plan-prompt pattern six-for-six locally — until ornith6's double view-decimals failure breaks it at 6 bit, a reminder that the pattern too is quant-dependent; and the wedge aborted the rest of rep 2 without touching the ranking, because the rule forbids single-run verdicts in either direction.

9 Recommendation

Load Qwen3-Coder-30B-A3B 8-bit (local30x8) as the resident coder: one server, thinking off, output cap 8192, prefix cache on, the qwen3 parsers and 8 GB cache budget of the round-7 launcher, path and tool-name hooks registered before the first spawn. Its evidence: 5 of 6 on the three-task set, log-size 4 of 4 across both repetitions, and the head-to-head won on both axes at 3.8–4.4× ornith8's walls, at 17–18 GB resident in round 6 and 30.9–32.4 GB in the round-7 window — plan for the higher figure. Give it either prompt on one-function fixes and the precise prompt first on view-decimals and box-press, with the plan-arm sprawl noted. The seat carries one condition: rank 1 is diff-verified, not build-verified, so build-verify the committed log-size pass (`bd3ee3c0`-equivalent on the real fix) at the next daylight slot before calling the deal final.

Ornith-1.5-35B-A3B 8-bit drops to the fallback seat and keeps a real edge: its rep-1 passes are the night's only build-verified ones. If the daylight build fails local30x8's committed diff, the resident role reverts to ornith8 (one server, fold-text launcher, thinking off, output cap 8192, prefix cache on, 36 GB watch line): plan prompt on view-decimals-class tasks, either prompt on one-function fixes, box-press with the precise prompt first.

Keep the save-model-ref class and any multi-file, reference-preserving change on the cloud coder (Sonnet 5.5); no local model has crossed it in 14 graded tries, and the class now carries a second strike: it is also the wedge trigger. Keep qwen38 as the light fallback (∼20 GB) with its record now complete (3 of 8) and its dense decode visibly slower than the MoE arms under two parallel agents; do not adopt glimmer or gpt-oss-20b on this stack. Do not serve models larger than memory as workers until the prefix-cache and paging questions are solved — treat ddalcu's 3 of 6 as the upper bound of what SSD paging gives for free. For the next run: the

daylight build-verification of local30x8's committed pass first; ornith8's rep-2 completion on the remaining three tasks is optional; add the Metal-peak watch line.

10 Limitations

The ranking stands on rep-1 evidence with one replicated cell plus the head-to-head's log-size rep 2; 5 of 8 and 7 of 17 are small fractions of small denominators, and the Wilson intervals overlap for the middle ranks. Ranks 1–4 rest on buildless diff grading: rank 1's passes — including both head-to-head arms — are inspected checks containing the true fixes, not green test suites, so the seat is diff-verified rather than build-verified, and the runner-up keeps the only build-verified passes. One of the diff-graded rows (codernext log-size) self-reported failure because it could not build — the fraction is a lower bound on capability and an upper bound on verified quality. The two round-7 candidates were rejected before serving (model format; no quant fits free RAM) and are therefore unmeasured, not failed. The 4-bit arm's 17 graded arms were collected under night-2's serving, so the A/B compares bit widths across nights — launcher and cache fixes are common to both, but not every serving detail is identical; the night-1 4-bit local30 comparison crosses harness versions as well. The deadline structure differed across the queue (12–30 min caps), so per-task fractions are comparable within a candidate and only roughly across candidates. Glimmer's voids reduce its graded record to five arms, all fails, which is enough to reject it as the resident but not to characterise it; two ornith6 one-turn arms were not inspected and their class assignment is provisional. The wedge was measured twice in effect but instrumented once; its trigger conditions (long context, continuous batching, 35–45 minutes) are an observation, not an isolated cause. The tool-error log for the night also contains 175 rows from an out-of-round infrastructure session; they are excluded from every count here. Grading is mechanical, prompts and plans were written by cloud models, and a cloud model is also the yardstick. All samples come from one subsystem of one code base on one machine.

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